
RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Computer Science & Engineering

Numerical Methods Laboratory Manual

Interpolation

Chapter 3 · Finite Differences & Interpolation Formulae

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Topics: Finite Differences (forward, backward, central),
Newton's Forward & Backward Interpolation,
Gauss's Central Difference Formulae, Stirling's Formula,
Lagrange's Interpolation, and Inverse Interpolation

Reference: S.S. Sastry, *Introductory Methods of
Numerical Analysis*, 5th ed., PHI Learning

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1 Introduction and How to Use This Manual

This manual accompanies the lecture material on **Chapter 3: Interpolation**. Each experiment treats one *interpolation technique* and is organised uniformly:

- **Objective** — what the experiment teaches.
- **Theory** — a short statement of the method and the formula it uses.
- **Algorithm** — complete *pseudocode* you must implement.
- **Worked Example** — a fully solved example showing the *exact input* (the data table) and the *complete output* (difference table and interpolated value).
- **Exercise** — at least ten problems to be solved and reported.

Convention. For the equal-interval methods the nodes are $x_i = x_0 + ih$ with constant step h , and the interpolation variable is $p = \frac{x - x_0}{h}$ (taken about the relevant anchor node). A **difference table** is always constructed *before* an equal-interval formula is applied. Lagrange's formula and inverse interpolation allow *arbitrary* spacing and need no difference table. Tabulate the full difference table and report interpolated values to the accuracy warranted by the data.

A general-purpose driver (in C/C++/Python/MATLAB) is recommended: read the data table (x_i, y_i) and the value x at which the function is wanted, build the difference table where required, then evaluate the chosen formula and print both the table and the interpolated value.

2 Lab Report: Format and Submission Guidelines

Every submitted report **must** follow the structure and rules below.

2.1 Report structure

1. **Cover page** — Course title and code, Experiment number and name, your full name, roll/registration number, section, group, date of performance, date of submission.
2. **Objective** — one or two lines.
3. **Theory** — the method, its formula and where it applies (in your own words; 4–6 lines).
4. **Algorithm / Flowchart** — the pseudocode (or a flowchart) you implemented.
5. **Source code** — complete, commented program listing.
6. **Sample input** — the data table and the value(s) of x used.
7. **Output / Results** — the *full difference table* (for equal-interval methods) and the interpolated value. Paste the program output *and* present a neat hand-formatted table.
8. **Discussion** — which formula suited the position of x , the order at which differences became negligible, the effect of degree, and any difficulty faced.
9. **Conclusion** — 2–3 lines.

2.2 Formatting rules

- Paper size **A4**; margins **2.5 cm** on all sides.
- Body font **12 pt** serif (Times/Latin Modern); line spacing **1.5**; justified text.
- Headings bold; code in a **monospaced** font (e.g. Consolas/Courier), 10–11 pt, with line numbers.
- Every table and figure must be **numbered and captioned**. Tabulate difference data with aligned decimal points.
- Number all equations you reference.

2.3 Submission rules

- Submit a single **PDF** named `Roll_ExpNo_MethodName.pdf` — for example, `2303001_Exp06_Lagrange-Interpolation.pdf`.
- Submit the **source files** separately in a folder/zip with the same base name.
- Deadline: **within two weeks** of performing the experiment unless announced otherwise. Late reports lose marks as per course policy.
- Work must be **your own**. Copied reports/code receive zero; both provider and copier are penalised. You may discuss ideas but must write and code independently.

Marking (typical). Algorithm & code correctness 40%, results & difference tables 25%, discussion 20%, formatting & neatness 10%, viva/demo 5%.

3 Experiment 1: Finite Difference Tables

Objective

To construct the forward (Δ), backward (∇) and central (δ) difference tables of a tabulated function, to read the leading differences, and to use the symbolic operators (E , μ , D) to detect the polynomial degree, fill a missing entry and prove difference identities.

3.1 Theory

For equally spaced data $x_i = x_0 + ih$ the *forward*, *backward* and *central* differences are

$$\Delta y_i = y_{i+1} - y_i, \quad \nabla y_i = y_i - y_{i-1}, \quad \delta y_i = y_{i+1/2} - y_{i-1/2}.$$

With the *shift operator* $Ey_i = y_{i+1}$ they satisfy the operator identities

$$\Delta \equiv E - 1, \quad \nabla \equiv 1 - E^{-1}, \quad \delta \equiv E^{1/2} - E^{-1/2}, \quad E \equiv e^{hD}.$$

A degree- m polynomial has $\Delta^m y$ constant and $\Delta^{m+1} y = 0$; this lets one judge the degree and recover a missing value.

Algorithm 1 Build the Forward Difference Table

Input: ordinates $y[0..n]$ (equally spaced, step h)

Output: table $D[k][i] = \Delta^k y_i$

```

1: for  $i = 0$  to  $n$  do
2:    $D[0][i] \leftarrow y[i]$ 
3: end for
4: for  $k = 1$  to  $n$  do
5:   for  $i = 0$  to  $n - k$  do
6:      $D[k][i] \leftarrow D[k-1][i+1] - D[k-1][i]$ 
7:   end for
8: end for
9: return  $D$ 
```

$\triangleright D[k][0] = \Delta^k y_0$ are the leading differences

3.2 Worked Example (full input and output)

Example 1.1 — forward difference table of $y = x^3$

Input. $y = x^3$ at $x = 0, 1, 2, 3, 4$ (so $h = 1$).

Output.

x	$y = x^3$	Δ	Δ^2	Δ^3	Δ^4
0	1				
		7			
1	8		12		
		19		6	
2	27		18		0
		37		6	
3	64		24		
		61			
4	125				

The third differences are **constant** ($\Delta^3 y = 6$) and $\Delta^4 y = 0$ — the signature of a cubic. The leading differences are $\Delta y_0 = 7$, $\Delta^2 y_0 = 12$, $\Delta^3 y_0 = 6$.

Example 1.2 — a missing tabular value

Input. $x = 0, 1, 2, 3, 4$, $y = 1, 3, 9, \square, 81$ (the value at $x = 3$ is lost); the data are known to follow a cubic, so $\Delta^4 y_0 = 0$.

Output. $\Delta^4 y_0 = y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$ gives $81 - 4y_3 + 6(9) - 4(3) + 1 = 0$, hence $y_3 = 31$. (The true function is 3^x , for which $3^3 = 27$; the cubic fit differs because 3^x is not a polynomial.)

Exercise 1 (Finite Differences)

1. Construct the forward difference table for $f(x) = x^3 - 3x^2 + 5x - 7$ at $x = -1, 0, 1, 2, 3, 4$; state $\Delta^3 f$ and $\Delta^4 f$.
2. Fit a polynomial to $x = 0, 1, 2, 3, 4$, $y = 1, 3, 9, \square, 81$; using $\Delta^4 y_0 = 0$ find the missing term.
3. The population (thousands) at the censuses 1951–1991 (every 10 yr) was 35, 42, 58, 84, 120. Build the difference table and state the lowest order at which the differences become roughly constant.
4. Prove $\Delta^n y_0 = \sum_{r=0}^n (-1)^r \binom{n}{r} y_{n-r}$ by separation of symbols.
5. Show $E \equiv 1 + \Delta \equiv e^{hD}$, hence $hD = \log(1 + \Delta) = \Delta - \frac{1}{2}\Delta^2 + \frac{1}{3}\Delta^3 - \dots$.
6. Evaluate (with $h = 1$): (a) $\Delta(e^{ax})$, (b) $\Delta^2(2x^3 - 3x^2 + 1)$, (c) $\Delta\left(\frac{x}{x+1}\right)$.
7. Prove $\delta = \Delta E^{-1/2} = \nabla E^{1/2}$ and $\mu^2 = 1 + \frac{1}{4}\delta^2$.
8. One value in $u_0, \dots, u_6 = 25, 35, 52, 70, 91, 116, 145$ is wrong. Build the table and use the binomial spread of the error through Δ^3, Δ^4 to locate and correct it.
9. Express $\nabla^3 y_5$ and $\delta^4 y_2$ in terms of the ordinates y_i .
10. Construct the forward, backward *and* central difference tables for $x = 1, 2, 3, 4, 5$, $y = 1, 16, 81, 256, 625$ ($= x^4$) and verify that the same numbers occupy the same positions.
11. By separation of symbols prove $u_x = u_0 + \binom{x}{1}\Delta u_0 + \binom{x}{2}\Delta^2 u_0 + \dots$.
12. Given $y_0, \dots, y_4$, write $\nabla^4 y_4$ and $\Delta^2 y_1$ explicitly and verify $\Delta^2 y_1 = \nabla^2 y_3 = \delta^2 y_2$ on the data of problem 10.

4 Experiment 2: Newton's Forward Difference Interpolation

Objective

To estimate y at a point x lying *near the beginning* of an equally-spaced table using Newton's forward difference formula.

4.1 Theory

With $p = \frac{x - x_0}{h}$ and the leading (top-diagonal) forward differences,

$$y_p = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots \quad (3.9)$$

It is most accurate when x is near x_0 (so p is small and positive). The error after the n th difference is $\frac{p(p-1) \cdots (p-n)}{(n+1)!} h^{n+1} y^{(n+1)}(\xi)$.

Algorithm 2 Newton's Forward Difference Interpolation

Input: table $x[0..n]$, $y[0..n]$ (step h); value X

Output: interpolated value y_p

- 1: build forward difference table; $\Delta^k y_0 \leftarrow D[k][0]$
- 2: $p \leftarrow (X - x[0])/h$
- 3: $y_p \leftarrow y[0]$; $term \leftarrow 1$
- 4: **for** $k = 1$ **to** n **do**
- 5: $term \leftarrow term \cdot (p - (k - 1))/k$
- 6: $y_p \leftarrow y_p + term \cdot \Delta^k y_0$
- 7: **end for**
- 8: **return** y_p

4.2 Worked Example (full input and output)

Example 2.1 — a cubic through four points

Input. $y(1) = 24$, $y(3) = 120$, $y(5) = 336$, $y(7) = 720$; here $h = 2$, $x_0 = 1$, $p = \frac{x-1}{2}$.

Output. Difference table:

x	y	Δ	Δ^2	Δ^3
1	24			
		96		
3	120		120	
		216		48
5	336		168	
		384		
7	720			

Leading differences $\Delta y_0 = 96$, $\Delta^2 y_0 = 120$, $\Delta^3 y_0 = 48$ ($\Delta^4 y_0 = 0$). Substituting in (3.9),

$$y(x) = 24 + \frac{x-1}{2}(96) + \frac{(\frac{x-1}{2})(\frac{x-1}{2}-1)}{2}(120) + \frac{(\frac{x-1}{2})(\frac{x-1}{2}-1)(\frac{x-1}{2}-2)}{6}(48) = x^3 + 6x^2 + 11x + 6.$$

At $x = 8$, $p = \frac{7}{2}$ gives $y(8) = 990$ (exact, since the data are a cubic — interpolation *and* extrapolation are exact for polynomials).

Example 2.2 — sum of cubes

Input. $S_n = 1^3 + 2^3 + \cdots + n^3$; since $\Delta S_n = (n+1)^3$ the leading differences are $S_1 = 1$, $\Delta S_1 = 8$, $\Delta^2 S_1 = 19$, $\Delta^3 S_1 = 18$, $\Delta^4 S_1 = 6$.

Output. Newton's forward formula (replacing p by $n-1$) gives

$$S_n = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2 = \left[\frac{n(n+1)}{2} \right]^2.$$

Exercise 2 (Newton Forward)

1. For $(1, 24), (3, 120), (5, 336), (7, 720)$ ($h = 2$) find $y(2)$.
2. From $y = x^3$ at $x = 1, 2, 3, 4, 5$, estimate $y(1.5)$.
3. f at $x = 40, 50, 60, 70, 80, 90$ is 184, 204, 226, 250, 276, 304; find $f(43)$.
4. e^x at $x = 1.0(0.1)1.4$ is 2.7183, 3.0042, 3.3201, 3.6693, 4.0552; estimate $e^{1.05}$.
5. The population (thousands) in 1891–1931 (every 10 yr) is 46, 66, 81, 93, 101; estimate the population in 1895.
6. $\tan x$ at $x = 0.10(0.05)0.30$ is 0.1003, 0.1511, 0.2027, 0.2553, 0.3093; find $\tan 0.12$.
7. From $\sin \theta$ at $\theta = 0^\circ, 10^\circ, 20^\circ, 30^\circ, 40^\circ$, estimate $\sin 5^\circ$.
8. Derive $\sum_{k=1}^n k^2$ as a polynomial in n by Newton's forward formula.
9. Given $f(0) = 1, f(1) = 0, f(2) = 1, f(3) = 10$, find the interpolating cubic and $f(0.5)$.
10. $\log_{10} x$ at $x = 1, 2, 3, 4, 5$ is 0, 0.3010, 0.4771, 0.6021, 0.6990; estimate $\log_{10} 1.5$ and comment on the accuracy (why is it only fair?).
11. The density of water (g/cm^3) at $T = 0, 5, 10, 15, 20^\circ\text{C}$ is 0.99987, 0.99999, 0.99973, 0.99913, 0.99823; estimate the density at 3°C .
12. From y at $x = 0, 1, 2, 3, 4, 5$ equal to 0, 1, 8, 27, 64, 125, build the table, write the polynomial, and confirm $\Delta^3 y$ is constant.

5 Experiment 3: Newton's Backward Difference Interpolation

Objective

To estimate y at a point x lying *near the end* of an equally-spaced table using Newton's backward difference formula.

5.1 Theory

Anchoring at the last node x_n with $p = \frac{x - x_n}{h}$ (so $p < 0$ when interpolating) and the bottom-diagonal backward differences,

$$y_p = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \cdots + \frac{p(p+1) \cdots (p+n-1)}{n!} \nabla^n y_n \quad (3.14)$$

It is most accurate when x is near x_n .

Algorithm 3 Newton's Backward Difference Interpolation

Input: table $x[0..n]$, $y[0..n]$ (step h); value X

Output: interpolated value y_p

- 1: build difference table; $\nabla^k y_n \leftarrow D[k][n-k]$
 - 2: $p \leftarrow (X - x[n])/h$
 - 3: $y_p \leftarrow y[n]$; $term \leftarrow 1$
 - 4: **for** $k = 1$ **to** n **do**
 - 5: $term \leftarrow term \cdot (p + (k-1))/k$
 - 6: $y_p \leftarrow y_p + term \cdot \nabla^k y_n$
 - 7: **end for**
 - 8: **return** y_p
-

5.2 Worked Example (full input and output)

Example 3.1 — $\sin 38^\circ$ from a table of sines

Input. $\sin x$ for $x = 15^\circ(5^\circ)40^\circ$, step $h = 5$. 38° is near the end, so anchor at $x_n = 40^\circ$,
 $p = \frac{38 - 40}{5} = -0.4$.

Output. The bottom (backward) diagonal is

$$\nabla y_{40} = 0.0692112, \nabla^2 = -0.0043652, \nabla^3 = -0.0005599, \nabla^4 = 0.0000289, \nabla^5 = 0.0000041.$$

Substituting in (3.14) with $p = -0.4$,

$$\sin 38^\circ = 0.6427876 - 0.0276845 + 0.0005238 + 0.0000358 - 0.0000012 = \boxed{0.6156615},$$

agreeing with the true value 0.6156615 to seven decimals.

Exercise 3 (Newton Backward)

1. From $\sin x$ at $x = 15^\circ(5^\circ)40^\circ$, find $\sin 38^\circ$.
2. f at $x = 40, 50, 60, 70, 80, 90$ (184, 204, 226, 250, 276, 304); estimate $f(85)$.
3. e^x at $x = 1.0(0.1)1.4$; estimate $e^{1.38}$.
4. Population 46, 66, 81, 93, 101 (1891–1931); estimate the population in 1925.
5. $\tan x$ at $0.10(0.05)0.30$; estimate $\tan 0.28$.
6. From $y = x^3$ at $x = 1, 2, 3, 4, 5$, estimate $y(4.5)$ by the backward formula.
7. $\cos \theta$ at $\theta = 0^\circ(10^\circ)60^\circ$; estimate $\cos 55^\circ$.
8. Given $f(1) = 1, f(2) = 8, f(3) = 27, f(4) = 64, f(5) = 125$, find $f(4.7)$.
9. $\log_{10} x$ at $x = 300, 310, 320, 330, 340$ is 2.4771, 2.4914, 2.5051, 2.5185, 2.5315; estimate $\log_{10} 337$.
10. The pressure of saturated steam (kPa) at $T = 100, 110, 120, 130, 140^\circ\text{C}$ is 101.3, 143.3, 198.5, 270.1, 361.3; estimate the pressure at 138°C .
11. Show that the backward formula reproduces every tabulated value exactly at its own node.
12. The yearly sales (lakh \$) for 2016–2021 are 12, 15, 20, 27, 39, 52; estimate the sales in 2020.5 (mid-year).

6 Experiment 4: Gauss's Central Difference Interpolation

Objective

To estimate y at a point lying *near the middle* of an equally-spaced table using Gauss's forward and backward central-difference formulae, which use the differences symmetrically about a central ordinate y_0 .

6.1 Theory

Label the node nearest x as the central ordinate y_0 and put $p = \frac{x - x_0}{h}$. Then Gauss's *forward* formula (3.16) is

$$y_p = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!} \Delta^3 y_{-1} \\ + \frac{(p+1)p(p-1)(p-2)}{4!} \Delta^4 y_{-2} + \dots$$

and Gauss's *backward* formula (3.19) is

$$y_p = y_0 + p \Delta y_{-1} + \frac{p(p+1)}{2!} \Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!} \Delta^3 y_{-2} \\ + \frac{(p+2)(p+1)p(p-1)}{4!} \Delta^4 y_{-2} + \dots$$

The forward form is used for $0 < p < 1$, the backward form for $-1 < p < 0$.

Algorithm 4 Gauss's Forward Central Difference Interpolation

Input: central difference table; central index 0 (so $y_{-m}, \dots, y_m$); value X

Output: interpolated value y_p

- 1: $p \leftarrow (X - x_0)/h$
 - 2: $y_p \leftarrow y_0 + p \cdot \Delta y_0$
 - 3: $y_p \leftarrow y_p + \frac{p(p-1)}{2!} \Delta^2 y_{-1}$
 - 4: $y_p \leftarrow y_p + \frac{(p+1)p(p-1)}{3!} \Delta^3 y_{-1}$
 - 5: $y_p \leftarrow y_p + \frac{(p+1)p(p-1)(p-2)}{4!} \Delta^4 y_{-2}$
 - 6: ... (continue with the central zig-zag while the differences matter)
 - 7: **return** y_p
-

6.2 Worked Example (full input and output)

Example 4.1 — $e^{1.17}$ by Gauss's forward formula

Input. e^x for $x = 1.00(0.05)1.30$. Take the central node $x_0 = 1.15$, so $p = \frac{1.17 - 1.15}{0.05} = 0.4$.

Output. Central difference table (central row in bold):

x	e^x	Δ	Δ^2	Δ^3	Δ^4
1.00	2.7183				
		0.1394			
1.05	2.8577		0.0071		
		0.1465		0.0004	
1.10	3.0042		0.0075		0
		0.1540		0.0004	
1.15	3.1582		0.0079		0
		0.1619		0.0004	
1.20	3.3201		0.0083		0.0001
		0.1702		0.0005	
1.25	3.4903		0.0088		
		0.1790			
1.30	3.6693				

With $y_0 = 3.1582$, $\Delta y_0 = 0.1619$, $\Delta^2 y_{-1} = 0.0079$, $\Delta^3 y_{-1} = 0.0004$ and $p = 0.4$,

$$e^{1.17} = 3.1582 + 0.0648 - 0.0009 - 0.0000 = \boxed{3.2221},$$

matching the true value $3.22199\dots$ using only first and second central differences.

Exercise 4 (Gauss Central)

1. e^x at $1.00(0.05)1.30$; find $e^{1.17}$ (central node 1.15, $p = 0.4$).
2. From $\sin \theta$ at $\theta = 20^\circ(10^\circ)60^\circ$, find $\sin 35^\circ$ by Gauss's forward formula.
3. y at $x = 0, 1, 2, 3, 4, 5, 6$ equal to $1, 8, 27, 64, 125, 216, 343$; find $y(3.3)$.
4. $\log_{10} x$ at $x = 1.0(0.5)4.0$ ($0, 0.176, 0.301, 0.477, 0.602, 0.699, 0.778$); estimate $\log_{10} 2.6$.
5. $\tan x$ at $x = 0.10(0.05)0.30$; find $\tan 0.21$ by a central formula.
6. Estimate $f(0.25)$ from f at $x = -2, -1, 0, 1, 2$ equal to $2.626, 0.342, 0, 0.342, 2.626$ (central node 0).
7. For the table of problem 1, compute $e^{1.17}$ by Gauss's *backward* formula about $x_0 = 1.20$ ($p = -0.6$) and confirm the two agree.
8. From e^{-x} at $x = 0.0(0.1)0.6$, estimate $e^{-0.27}$.
9. From a 7-point table, interpolate at $p = 0.3$ about the centre and report how many central differences were needed for 4-figure accuracy.
10. From the calibration table (input $\rightarrow$ reading) $(10, 2.1), (20, 4.0), (30, 5.6), (40, 6.9), (50, 8.0)$, find the reading at input 33.
11. $\cosh x$ at $x = 0.5(0.02)0.58$ ($1.1276, 1.1417, 1.1561, 1.1707, 1.1855$); estimate $\cosh 0.552$.
12. Tabulate the central zig-zag used by Gauss forward and by Gauss backward for a 5-point set and mark the shared central second difference.

7 Experiment 5: Stirling's Central Difference Interpolation

Objective

To estimate y at a point *very close* to a central node by Stirling's formula — the mean of Gauss's forward and backward formulae — which is most accurate for $-\frac{1}{4} \leq p \leq \frac{1}{4}$.

7.1 Theory

Averaging Gauss's two formulae,

$$y_p = y_0 + p \frac{\Delta y_{-1} + \Delta y_0}{2} + \frac{p^2}{2!} \Delta^2 y_{-1} + \frac{p(p^2 - 1)}{3!} \frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} + \frac{p^2(p^2 - 1)}{4!} \Delta^4 y_{-2} + \dots \quad (3.21)$$

It averages the two adjacent first (and third) differences and is symmetric about y_0 .

Algorithm 5 Stirling's Central Difference Interpolation

Input: central difference table; central index 0; value X (with $|p| \leq \frac{1}{4}$)

Output: interpolated value y_p

- 1: $p \leftarrow (X - x_0)/h$
- 2: $y_p \leftarrow y_0 + p \cdot \frac{\Delta y_{-1} + \Delta y_0}{2}$
- 3: $y_p \leftarrow y_p + \frac{p^2}{2!} \Delta^2 y_{-1}$
- 4: $y_p \leftarrow y_p + \frac{p(p^2 - 1)}{3!} \cdot \frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2}$
- 5: $y_p \leftarrow y_p + \frac{p^2(p^2 - 1)}{4!} \Delta^4 y_{-2}$
- 6: **return** y_p

7.2 Worked Example (full input and output)

Example 5.1 — $e^{0.644}$ by Stirling's formula

Input. e^x for $x = 0.61(0.01)0.67$. Central node $x_0 = 0.64$, $p = \frac{0.644 - 0.64}{0.01} = 0.4$.

Output. The needed central differences are

$$y_0 = 1.896481, \quad \Delta y_{-1} = 0.018871, \quad \Delta y_0 = 0.019060, \quad \Delta^2 y_{-1} = 0.000189.$$

Third differences are negligible, so keeping two central terms,

$$e^{0.644} = 1.896481 + 0.4 \left(\frac{0.018871 + 0.019060}{2} \right) + \frac{(0.4)^2}{2} (0.000189) = \boxed{1.904082},$$

exact to six decimals.

Exercise 5 (Stirling)

1. e^x at $0.61(0.01)0.67$; find $e^{0.644}$ (central node 0.64).
2. $\cos \theta$ at $\theta = 0^\circ(10^\circ)60^\circ$ is $1.00000, 0.98481, 0.93969, 0.86603, 0.76604, 0.64279, 0.50000$; find $\cos 25^\circ$ (central node 30°).
3. From $\sin x$ at $x = 0^\circ(10^\circ)40^\circ$, find $\sin 16^\circ$.
4. y at $x = 20, 21, 22, 23, 24$ equal to $0.342, 0.358, 0.374, 0.391, 0.407$; estimate $y(22.4)$.
5. For a value with p near 0 , compute the estimate by Stirling and by Bessel and compare.
6. $\tan x$ at $x = 0.20(0.02)0.28$; estimate $\tan 0.245$.
7. From $\log_{10} x$ at $x = 2.0(0.1)2.4$, estimate $\log_{10} 2.21$ and compare the error of Newton-forward vs Stirling.
8. Estimate $f(0.5)$ from f at $x = -2, -1, 0, 1, 2$ equal to $0.135, 0.368, 1.000, 2.718, 7.389$ ($= e^x$).
9. Verify numerically that Stirling's formula equals the mean of Gauss's forward and backward formulae for the table of problem 1.
10. (Application) A thermocouple gives emf $0.798, 1.203, 1.612, 2.023, 2.436$ mV at $T = 20^\circ, 30^\circ, 40^\circ, 50^\circ, 60^\circ\text{C}$ respectively; estimate the emf at 42°C .
11. Use Stirling's formula to estimate both y and $y'(x_0)$ at the central node from a 5-point table.
12. From $J_0(x)$ at $x = 1.0(0.1)1.4$ ($0.7652, 0.7196, 0.6711, 0.6201, 0.5669$), estimate $J_0(1.22)$.

8 Experiment 6: Lagrange's Interpolation Formula

Objective

To fit a single polynomial through data at *arbitrarily* (unequally) spaced nodes and evaluate it at a required point, using Lagrange's interpolation formula — no difference table and no constant step h .

8.1 Theory

For $(n + 1)$ nodes (x_i, y_i) ,

$$L_n(x) = \sum_{i=0}^n l_i(x) y_i, \quad l_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}, \quad (3.38-3.39)$$

where the basis polynomials satisfy $l_i(x_j) = \delta_{ij}$ and $\sum_i l_i(x) = 1$. The numerator of l_i omits the factor $(x - x_i)$; the denominator is the same product evaluated at x_i .

Algorithm 6 Lagrange's Interpolation

Input: nodes $x[0..n]$, $y[0..n]$ (need not be equally spaced); value X

Output: interpolated value sum

```

1:  $sum \leftarrow 0$ 
2: for  $i = 0$  to  $n$  do
3:    $L \leftarrow 1$ 
4:   for  $j = 0$  to  $n$ ,  $j \neq i$  do
5:      $L \leftarrow L \cdot (X - x[j]) / (x[i] - x[j])$ 
6:   end for
7:    $sum \leftarrow sum + L \cdot y[i]$ 
8: end for
9: return  $sum$ 
```

8.2 Worked Example (full input and output)

Example 6.1 — $\log_{10} 301$ from four points

Input. $\log_{10} x$ at the unequally-spaced nodes $x = 300, 304, 305, 307$ with $y = 2.4771, 2.4829, 2.4843, 2.4871$; find $\log_{10} 301$.

Output. At $x = 301$ the factors $(x - x_i)$ are $1, -3, -4, -6$:

$$\begin{aligned}
l_0 y_0 &= \frac{(-3)(-4)(-6)}{(-4)(-5)(-7)} (2.4771) = 1.2739, & l_1 y_1 &= \frac{(1)(-4)(-6)}{(4)(-1)(-3)} (2.4829) = 4.9658, \\
l_2 y_2 &= \frac{(1)(-3)(-6)}{(5)(1)(-2)} (2.4843) = -4.4717, & l_3 y_3 &= \frac{(1)(-3)(-4)}{(7)(3)(2)} (2.4871) = 0.7106.
\end{aligned}$$

Summing, $\log_{10} 301 = 1.2739 + 4.9658 - 4.4717 + 0.7106 = \boxed{2.4786}$.

Example 6.2 — a quadratic for $\ln x$, then $\ln 2.7$

Input. $\ln x$ at $x = 2.0, 2.5, 3.0$ with $y = 0.69315, 0.91629, 1.09861$.

Output. The basis polynomials are $l_0 = 2x^2 - 11x + 15$, $l_1 = -(4x^2 - 20x + 24)$, $l_2 = 2x^2 - 9x + 10$, so

$$L_2(x) = -0.08164x^2 + 0.81366x - 0.60761.$$

At $x = 2.7$, $\ln 2.7 \approx L_2(2.7) = \boxed{0.9941164}$; the true value is 0.9932518, so $|\text{Error}| = 0.0008646$.

Exercise 6 (Lagrange)

1. $\log_{10} x$ at 300, 304, 305, 307 (2.4771, 2.4829, 2.4843, 2.4871); find $\log_{10} 301$.
2. $\ln x$ at 2.0, 2.5, 3.0; find the quadratic and $\ln 2.7$ with its error.
3. $\sin x$ at $0, \pi/4, \pi/2$ (0, 0.70711, 1.0); estimate $\sin(\pi/6)$.
4. Given (0, 2), (1, 3), (2, 12), (5, 147), find $y(x)$ and hence $y(3)$.
5. (5, 12), (6, 13), (9, 14), (11, 16); find $y(10)$.
6. (1, 1), (2, 8), (4, 64), (7, 343) (data of x^3 , unequal spacing); find $y(3)$.
7. From (0, -12), (1, 0), (3, 12), (4, 24) recover the polynomial $y(x)$.
8. For the nodes 1, 2, 4, write the three basis polynomials $l_i(x)$ and verify $\sum_i l_i(x) = 1$ and $l_i(x_j) = \delta_{ij}$.
9. Given $f(0) = 1, f(1) = 3, f(3) = 55, f(4) = 99$, fit a Lagrange cubic and estimate $f(2)$.
10. Show that Lagrange interpolation and Newton's divided-difference formula give the *same* polynomial for the data of problem 4.
11. (Application) A thermistor reads resistance R ($\text{k}\Omega$) at temperatures T : (0, 27.3), (25, 10.0), (50, 4.16), (100, 0.97); estimate T when $R = 6.0$ (interpolate T as a function of R).
12. Estimate $y(2.5)$ from (0, 0), (1, 1), (2, 16), (3, 81) ($= x^4$) and compare with the exact value; comment on the error of a cubic fit to a quartic.

9 Experiment 7: Inverse Interpolation by Lagrange's Formula

Objective

To find the value of x corresponding to a given y (the inverse problem) by interchanging the roles of x and y in Lagrange's formula.

9.1 Theory

Since Lagrange's formula does not require equal spacing in either variable, the abscissa x for a prescribed ordinate y is obtained by swapping x and y :

$$x = \sum_{i=0}^n \left(\prod_{\substack{j=0 \\ j \neq i}}^n \frac{y - y_j}{y_i - y_j} \right) x_i \quad (3.44)$$

The y_i must be *monotonic* over the chosen nodes for the inverse to be single-valued.

Algorithm 7 Inverse Interpolation (Lagrange)

Input: nodes $x[0..n]$, $y[0..n]$ (y monotonic); target Y

Output: estimated abscissa sum

```

1:  $sum \leftarrow 0$ 
2: for  $i = 0$  to  $n$  do
3:    $L \leftarrow 1$ 
4:   for  $j = 0$  to  $n$ ,  $j \neq i$  do
5:      $L \leftarrow L \cdot (Y - y[j]) / (y[i] - y[j])$ 
6:   end for
7:    $sum \leftarrow sum + L \cdot x[i]$ 
8: end for
9: return  $sum$ 
```

▷ x at which the function equals Y

9.2 Worked Example (full input and output)

Example 7.1 — find x when $y = 7$

Input. Nodes $(x, y) = (1, 4), (3, 12), (4, 19)$ (the data come from $y = x^2 + 3$); find x for which $y = 7$.

Output. Interchanging x and y ,

$$\begin{aligned}
 x &= \frac{(7-12)(7-19)}{(4-12)(4-19)}(1) + \frac{(7-4)(7-19)}{(12-4)(12-19)}(3) + \frac{(7-4)(7-12)}{(19-4)(19-12)}(4) \\
 &= \frac{1}{2} + \frac{27}{14} - \frac{4}{7} = \boxed{1.86}.
 \end{aligned}$$

The true value is $x = \sqrt{7-3} = 2.0$; the three coarse points give 1.86.

Exercise 7 (Inverse Interpolation)

1. $(1, 4), (3, 12), (4, 19)$; find x where $y = 7$.
2. From $\sin x$ at $x = 20^\circ, 30^\circ, 40^\circ$ (0.342, 0.500, 0.643), find the angle whose sine is 0.45.
3. $y = x^2$ at $(2, 4), (4, 16), (6, 36)$; find x for $y = 12$ and compare with $\sqrt{12}$.
4. From e^x at $x = 1.0, 1.1, 1.2, 1.3$ (2.7183, 3.0042, 3.3201, 3.6693), find x such that $e^x = 3.0$.
5. Strain $y = 0.5x^2$ tabulated at $x = 1, 2, 3$ (0.5, 2.0, 4.5); find x where $y = 3.0$ and compare with $\sqrt{6}$.
6. From $\log_{10} x$ at $x = 2, 3, 4$ (0.3010, 0.4771, 0.6021), find x for which $\log_{10} x = 0.4$.
7. Given (x, y) near a sign change, use inverse interpolation to estimate the *root* (the x where $y = 0$): $(0.5, -0.18), (0.6, 0.05), (0.7, 0.30)$.
8. From a calibration table (input $x \rightarrow$ output y) $(10, 2.1), (20, 4.0), (30, 5.6), (40, 6.9)$, find the input giving output 5.0.
9. Compare inverse interpolation with solving $f(x) = Y$ by Newton–Raphson for the data of problem 4; comment on accuracy and effort.
10. From a thermocouple table emf(T): $(20, 0.798), (30, 1.203), (40, 1.612), (50, 2.023)$, find the temperature for a measured emf of 1.5 mV.
11. Find the abscissa of the maximum of f tabulated at $(1, 2), (2, 5), (3, 6), (4, 5), (5, 2)$ by inverse-interpolating $f'(x) = 0$.
12. For the data of problem 1, add the point $(0, 3)$ and recompute x at $y = 7$; comment on the improvement.

Appendix: Quick Comparison of the Methods

Method	Formula / use	Spacing	Best when
Difference table	$\Delta y_i = y_{i+1} - y_i, \nabla, \delta$	equal	foundation; find missing term / degree
Newton forward	$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \dots$	equal	x near the <i>start</i>
Newton backward	$y_p = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \dots$	equal	x near the <i>end</i>
Gauss central	symmetric zig-zag about y_0	equal	x near the <i>middle</i>
Stirling	mean of Gauss forward & backward	equal	x very near a node, $ p \leq \frac{1}{4}$
Lagrange	$L_n(x) = \sum_i l_i(x) y_i$	unequal	any spacing, one-off value
Inverse interp.	swap $x \leftrightarrow y$ in Lagrange	unequal	find x given y (y monotonic)

Choosing a formula. Same data, same $p = \frac{x - x_0}{h}$ — only the *stencil* changes. Use Newton forward/backward near the ends, Gauss/Stirling near the middle (they converge faster there because they use *symmetric* differences), and Lagrange whenever the abscissae are unequally spaced.

Reference: S.S. Sastry, *Introductory Methods of Numerical Analysis*, 5th ed., PHI Learning — § 3.1, 3.3, 3.6, 3.7.1–3.7.2, 3.9.1.